



leading technology and artificial intelligence (AI) data solutions firm.

Robots farmers?

Every industry that AI enters in a big way is also wary about how good the AI is. If it gets too good, it could just replace farmers. So is the inevitable future of farming just AI and robots doing it all? Maybe someday, however for now, it's focussed on aiding the farmer and making his business more efficient and more profitable.

"AI sensors can effectively ascertain the possibility of weed growth in the farms and also suggest the best herbicide as a solution. Many mechanical companies fashioned robots with spray guns to correctly watch out for weeds by using computer vision. These robots can slash the amount of pesticides sprayed on crops by 80% and the cost of herbicides by 90%. By substantially curtailing the quantity of pesticides required in the fields, these discreet AI sprayers can uplift the quality of agricultural output, while also bolstering economic efficiency," elaborates Saurabh Agarwal, founder and director of GROWiT India Pvt Ltd, India's first Direct-to-Farmer (D2F) protective farming agritech entity. The firm specialises in creating high-quality and cost-effective protective farming materials and manufacturing products, plus sets an optimal standard for the Indian agricultural and farming industry, ensuring heavier yields and lower carbon footprint.

With expertise in products like mulch films, crop cover, fruit cover, shade nets etc., protective farming has proved to be one of the finest agricultural rituals over time. Concepts like Taluka Mapping (agricultural region mapping) of India and AI-backed smart tools to calculate the ROI (Return on Investment), all benefit farmers.

When asked whether smart tractors and robots can replace farmers in the fields, Rashi Gupta feels that technology's will primarily heighten human capability.

"Smart equipment can ideally support farmers instead of removing them from the farmlands. It can

definitely share the manual workload, but a bulk of operations will still depend on the farmers only," concurs Rajesh Aggarwal. AI-propelled harvesting robots and driverless or smart tractors, he thinks, may address the issue of farm labor crisis. Nonetheless, these technologies remain in the initial stages of their implementation. Their large-scale use is yet to be seen in India.

The AI weatherman

Climate change is an important factor that makes weather predictions or monitoring critical to farmers. "The rise in erratic weather patterns and food insecurity challenges makes the modernization of agriculture imperative. We must also factor in the sharp spike in demand for food, thanks to the bulging global population bursting at the seams," reasons Navneet Ravikar, chairman and managing director, Leads Connect Services Pvt. Ltd., an agriculture and farm risk management company. Farmers are often dependent on monsoons for sowing crops. Can sporadic weather be predicted by AI to alert farmers to stay well-prepared for the most favorable season? "Yes, there has been a substantial development and accomplishment so far in the predictive analytics for construing the dynamics of weather behaviour. Space tech, backed by AI platforms, is now in a stage of issuing advisory warnings to farmers for bracing up against any untoward situation," reveals Ravikar.

Registering noticeable developments in the last few years in all aspects of the farming process, the agriculture sector in India exhibited immense

growth. Realising this, the Government of India unveiled a number of projects to obliterate challenges associated with cultivation. "However, what lingered as a big woe for the farmers is the hostile weather. It massively affects soil moisture, essential for the quality production of crops," says Mallory Dodd. She adds, "Using software solutions to model weather data against commonly-used AgTech algorithms could be beneficial to making short-term predictions in particular. That said, seasonal forecasts are much more complicated and would require not only the baseline data and AI technologies but also human expertise to make perfect predictions."

The most crucial aspect of AI in agriculture is that it is capable of interpreting weather conditions, allowing farmers to gear up in advance for planting crop seeds or saplings. "We also need to comprehend that AI can only provide trends and predictions. Everyone is aware that farmers are quite dependent on the monsoons for their crop yields. In that case, if farmers will be able to retrieve information about the rains precisely, it will



then not only prevent damage to the crops but also save their time and efforts. Weather-predicting technology piloted by AI will help farmers explore and plan when the crops can be sown or harvested," shares Aggarwal.

Economics

"The global agricultural market size is estimated to touch an astronomical figure of USD 7.22 billion by 2026," says Ravikar. Food is a huge industry in India. Agriculture alone is an important part of the Indian economy, contributing about 17% to the total GDP (Gross Domestic Product) and generating employment for over 60% of the popula-